

CAROLYN KROGER, PH.D.

Postdoctoral Research Fellow, Kresge Hearing Research Institute,
University of Michigan, Michigan Medicine
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EDUCATION

- Ph.D., Psychology** **Dec 2022**
- Michigan State University, East Lansing, MI Cognition and Cognitive Neuroscience Program
 - Dissertation: “Linking the spatial dimension to the timing control of rhythmic movements”
 - Advisor: J. Devin McAuley, Ph.D.
- M.A., Psychology** **May 2019**
- Michigan State University, East Lansing, MI Cognition and Cognitive Neuroscience Program
 - Thesis: “Revisiting the bimanual advantage in synchronization-continuation tapping”
 - Advisor: J. Devin McAuley, Ph.D.
- B.A., Psychology, *Magna Cum Laude*** **May 2015**
- University of Arkansas, Fayetteville, AR
 - Thesis: “The sound or the expectation: Psychological processes in the evaluation of musical performance”
 - Advisor: Elizabeth H. Margulis, Ph.D.
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ACADEMIC POSITIONS

- Postdoctoral Research Fellow** **Jan 2023 – Present**
University of Michigan
PI: Anahita Mehta – Auditory Perception and Neuroscience Lab
- NIH F32-funded study: Spatiotemporal interactions among sequential sounds in adults with cochlear implants
 - NIH T32-funded study: How changes in pitch bias timing judgments for auditory sequences
 - HHF ERG-funded study: Effects of healthy aging on perception of pitch, spatial, and timing features in auditory rhythms and speech
- Graduate Research Assistant** **August 2016 – Dec 2022**
Michigan State University
PI: J. Devin McAuley – Timing, Attention, and Perception Lab
- Interactions between temporal and spatial dimensions during rhythmic movements
 - Role of pitch and rhythm in speech perception in children and adults of varying language background
- Undergraduate Research Assistant** **Jan 2014 – July 2016**
University of Arkansas
PI: Elizabeth H. Margulis – Music Cognition Lab
- Researched effects of verbal contextual information on music perception
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GRANTS

Pending Proposals

NIH K99/R00 (PI – Kroger)

Submitted March 2026

National Institute of Deafness and Other Communication Disorders

“Dissociating Perceptual and Neural Correlates of Event-based and Emergent Auditory Rhythms”

Mentorship team: Ross K. Maddox, Anahita H. Mehta, Nicolaas I. Bohnen

Current Research

Hearing Health Foundation Emerging Research Grants (PI – Mehta)

Oct 1, 2024 – Oct 1, 2026

\$100,000

“Effects of age on interactions of acoustic features with timing judgements in auditory sequences”

Key Personnel

Completed Research

NIH F32 DC022162 (PI – Kroger)

Aug 1, 2024 – Dec 31, 2025

National Institute of Deafness and Other Communication Disorders

\$104,778

“Interactions between spatial and temporal cues for auditory grouping in normal hearing and single-sided deaf populations”

Principal Investigator 100% effort

NIH T32 DC000011 HBCS Postdoctoral Training Grant ()

Jan 2023 – Aug 2024

Student Undergraduate Research Fellowship

Jan 2015 – Aug 2015

Arkansas Department of Higher Education

HONORS & AWARDS

Association for Research in Otolaryngology – MidWinter Meeting Travel Award

Feb 2026

Acoustical Society of America –

Women in Acoustics Emerging Investigator Award

May 2025

Silas Hunt Distinguished Scholarship, *University of Arkansas*

Aug 2011 – May 2015

Arkansas Academic Challenge Scholarship, *ADHE*

Aug 2011 – May 2015

PUBLICATIONS

1. **Kroger, C.**, Fu, D. R., Banakis Hartl., R. M., and Mehta, A. H. (2026). A perceptual grouping model for auditory kappa effects. *Journal of Experimental Psychology: Human Perception and Performance*. <https://doi.org/10.1037/xhp0001433>
2. **Kroger, C.**, Fu, D. R., Banakis Hartl., R. M., and Mehta, A. H. (2024, preprint). Auditory kappa effects can be explained by perceptual grouping based on feature similarity. *PsyArXiv*
3. **Kroger, C.**, Kagerer, F. A., and McAuley, J. D. (2024). Interdependence of movement amplitude and tempo during self-paced finger tapping: Evaluation of a preferred velocity hypothesis. *Experimental Brain Research*, 1-12. <https://doi.org/10.1007/s00221-024-06814-x>
4. **Kroger, C.**, Kagerer, F. A., and McAuley, J. D. (2021). Monkey see, monkey tap: Mimicry of movement dynamics during coordinated tapping. *Experimental Brain Research*, 1-13. <https://doi.org/10.1007/s00221-021-06061-4>
5. **Kroger, C.** and Margulis, E. H. (2017). “But they told me it was professional:” Extrinsic factors in the evaluation of musical performance. *Psychology of Music*, 45(1), 49-64.

<https://doi.org/10.1177/0305735616642543>

6. Margulis, E. H., Levine, W. H., Simchy-Gross, R., and **Kroger, C.** (2017). Expressive intent, ambiguity, and aesthetic experiences of music and poetry. *PLoS One*. <https://doi.org/10.1371/journal.pone.0179145>
7. Jahanian Najafabadi, A., & **Kroger, C.** (2026). Altered gravity and time perception: a brief review. *Frontiers in Psychology*, 16, 1735072. <https://doi.org/10.3389/fpsyg.2025.1735072>
8. Jahanian Najafabadi, A., Botev, J., Sun, N., & **Kroger, C.** (2025). Virtual Tool Embodiment in Simulated Gravity Conditions. *Proceedings of the 17th International Workshop on Immersive Mixed and Virtual Environment Systems*, 65–71. <https://doi.org/10.1145/3712677.3720468>
9. Abdulrazzak, O., Ibrahim, N. I., Jones, G., Epperson, M. V., Onder, I., Graves, J., **Kroger, C.**, Mehta, A.H., and Hartl, R. M. B. (2025). Comparing bilateral and single-sided deaf cochlear implant recipients in a novel speech-in-noise and localization task. *Otolaryngology–Head and Neck Surgery*. <https://doi.org/10.1002/ohn.1187>

MANUSCRIPTS IN PROGRESS

1. **Kroger, C.**, Kagerer, F. A., and McAuley, J. D. (Under Review). Support for distinct timing control strategies during finger tapping: A spatiotemporal approach.
2. **Kroger, C.**, Sun, N., Botev, J., Kayser, C., Jahanian Najafabadi, A. (Under Review) Simulated Gravitational Physics and Time Perception in Virtual Reality.
3. Jahanian Najafabadi, A., Davoodi, E., **Kroger, C.** (In Preparation). Cortical activities are potential predictors of motor learning: A systematic review in healthy aging. Prospero Registration ID: CRD42024569699
4. **Kroger, C.**, Gordon, R. L., & McAuley, J. D. (In Preparation). Distal prosody affects novel artificial word learning in children.
5. **Kroger, C.**, Le, N. L., & McAuley, J. D. (In Preparation). Effects of distal prosody on novel word segmentation: A comparison of native and non-native English speakers.

CONFERENCE TALKS AND INVITED LECTURES

1. **Kroger, C.** (2025, September) Effects of pitch and space on auditory time perception in healthy aging. Hearing, Balance, and Chemical Senses (HBCS) Seminar, University of Michigan, Ann Arbor, MI.
2. **Kroger, C.** (2025, January) Influences of pitch and spatial trajectories on relative timing judgments in auditory sequences. Timing Research Forum. Invited lecture.
3. **Kroger, C.** (2024, November) Using auditory spatiotemporal interactions to evaluate spatial cue restoration with unilateral CI in single-sided deaf patients. CI CRASH conference, University of Wisconsin-Madison, Madison, WI.
4. **Kroger, C.** (2024, May) Influences of pitch and spatial trajectories on relative timing judgments in auditory sequences. Kresge Hearing Research Institute Founders Day, University of Michigan, Ann Arbor, MI.
5. **Kroger, C.** (2024, February) Influences of pitch and spatial trajectories on relative timing judgments in auditory sequences. Hearing, Balance, and Chemical Senses (HBCS) Seminar, University of Michigan, Ann Arbor, MI.
6. **Kroger, C.** (2023, May) Pitch contributions to timing judgments of auditory sequences. Kresge Hearing Research Institute Founders Day, University of Michigan, Ann Arbor, MI.

7. **Kroger, C.** (2022, November) Spatial contributions to auditory-motor synchronization. Auditory Perception, Cognition and Action Meeting (APCAM), Boston, MA.
8. **Kroger, C.** (2022, August) Putting the spatial dimension back into the investigation of rhythmic movement timing. Society for Music Perception and Cognition conference (SMPC), Portland, OR.
9. **Kroger, C.** (2022, February) Role of movement dynamics in timing performance during rhythmic tapping. Music Cognition Lab, Princeton University, Princeton, NJ. Invited lecture.
10. **Kroger, C.** (2022, February) Role of movement dynamics in timing performance during rhythmic tapping. Cognitive Science Forum, Michigan State University, East Lansing, MI.
11. **Kroger, C.** (2021, April). Monkey see, monkey tap: Mimicry of movement dynamics during coordinated tapping. New England Sequencing & Timing (NEST) conference, University of Connecticut, Storrs, CT.
12. **Kroger, C.** (2021, April). Are you a ‘bobber’ or a ‘dweller’? Individual differences in tapping to slow rhythms. Cognitive Science Forum, Michigan State University, East Lansing, MI.
13. **Kroger, C.** (2020, February) Monkey see, monkey tap: Mimicry of movement dynamics during coordinated tapping. Cognitive Science Forum, Michigan State University, East Lansing, MI.
14. **Kroger, C.** (2020, February) Monkey see, monkey tap: Mimicry of movement dynamics during coordinated tapping. Graduate Academic Conference (GAC), Michigan State University, East Lansing, MI.
15. **Kroger, C.** (2019, June). Revisiting the bimanual advantage in synchronization-continuation tapping. Rhythm Perception and Production Workshop (RPPW), Traverse City, MI.
16. **Kroger, C.** (2018, October). Cross-linguistic differences in the effects of distal prosody on speech segmentation. Tone and Intonation (TIE) Conference, University of Gothenburg, Gothenburg, SE.
17. **Kroger, C.** (2017, April). Rhythm perception in typical and atypical language development. Symposium on Timing and Rhythm (STaR), McMaster University, Hamilton, ON, Canada.
18. **Kroger, C.** (2015, August). The sound or the expectation: Psychological processes in the evaluation of musical performance. Society for Music Perception and Cognition conference (SMPC), Vanderbilt University, Nashville, TN.

CONFERENCE POSTERS

1. **Kroger, C.** (2026, February). Auditory spatio-temporal interactions in single-sided deaf patients with unilateral cochlear implants. Association for Research in Otolaryngology (ARO), San Juan, PR.
2. **Kroger, C.** (2025, October). Effects of non-temporal auditory features on timing judgments in healthy aging and cochlear implant users, Tokyo, JPN.
3. **Kroger, C.** (2025, May) Auditory spatio-temporal interactions in single-sided deaf patients with unilateral cochlear implants. Acoustical Society of America, New Orleans, LA.
4. **Kroger, C.** (2024, February) Influences of pitch and spatial trajectories on relative timing judgments in auditory sequences. Association for Research in Otolaryngology (ARO), Anaheim, CA.
5. **Kroger, C.** (2021, November). Are you a ‘bobber’ or a ‘dweller’? Individual differences in tapping to slow rhythms. Auditory Perception, Cognition, and Action Annual Meeting (APCAM) 2021, Online Conference.
6. **Kroger, C.** (2020, May). Investigating the use of distal prosody in speech segmentation in children.

- Expression, Language, and Music (ELM) conference, Hartford, CT. Poster accepted. (Conference cancelled).
7. **Kroger, C.** (2018, November). Cross-linguistic differences in the effects of distal prosody on speech segmentation. 59th annual meeting of the Psychonomic Society, New Orleans, LA.
 8. **Kroger, C.** (2018, November). Revisiting the bimanual advantage in synchronization- continuation tapping. Auditory Perception Cognition and Action Meeting (APCAM), New Orleans, LA.
 9. **Kroger, C., & Le, N.L.** (2018, September). Effects of distal prosody on novel word segmentation: A comparison of native and non-native English speakers. Generative Approaches to Language Acquisition North America (GALANA) Conference, Indiana University, Bloomington, IN.
 10. **Kroger, C., & Le, N.L.** (2018, April). Effects of distal prosody on novel word segmentation: A comparison of native and non-native English speakers. Cognitive Science of Communication Symposium (CSCS), Michigan State University, East Lansing, MI.
 11. **Kroger, C.** (2017, June). Rhythm perception in typical and atypical language development. The Neurosciences and Music VI Conference, Harvard Medical School, Boston, MA.

TEACHING AND MENTORING

Instructor of Record

Department of Psychology, Michigan State University	May – Aug 2021
• Course instructor PSY 200: Cognitive Psychology	
Department of Psychology, Michigan State University	May – Aug 2020
• Course instructor PSY 200: Cognitive Psychology	

Students Mentored

Graduate

• Zhiqing Yin, M.S., Current: PhD Student Case Western Reserve University	Jan – July 2023
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Post-baccalaureate

• Tess Starr, B.A.	May 2024 – Present
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Undergraduate

• Deborah Fu, Cognitive Science, University of Michigan	August 2023 – Present
• Mason Shields, Computer Science, University of Michigan	August 2023 – Present
• Sophia Riegler, National Technical Institute for the Deaf - RIT	May 2024 – August 2025
• Deshaun Omane, B.A., Chemistry, Macalester College	May – August 2023
• Anjelica Ferguson, B.A., Neuroscience, University of Michigan	May – August 2023
• Joscelyn Kooistra, B.A., Physics, University of Michigan	Jan 2023 – February 2024

Workshops and Courses Taken

University of Michigan Center for Research on Learning and Teaching	
• What Should I Do?: Supporting Students Experiencing Distress	March 2026
• Teaching Foundations: Strategies for Student Engagement	Feb 2026
• Metacognition: Strategies for Effective Learning	March 2025
• Teaching Foundations: Facilitating Discussion	September 2025

SERVICE & LEADERSHIP

UMPDA Board Member

June 2025 – Present

University of Michigan Postdoctoral Association

- Held office: Networking chair
- Organized and hosted regular networking and social events for postdoctoral fellows, including monthly coffee hours and happy hours and National Postdoc Appreciation Week.

ASA Administrative Committee Member

May 2025 – Present

Acoustical Society of America, Member Engagement Committee

- Organized and participated in events such as: Early-Career Acousticians Retreat, Affinity Groups and Listening Sessions, Early-Career Grant Panel, ASA Mentorship Program.
- Reviewed and scored applications for the Early Career Travel Award for the 2025 meeting in Honolulu, HI, and 2026 meeting in Philadelphia, PA.

MSU Cognitive Science Program Administrative Assistant

Aug 2019 – Dec 2022

Michigan State University Cognitive Science Program

- Organized semesterly Distinguished Speaker Series and faculty round-tables, webmaster for Cognitive Science Program website, aided with instruction and record-keeping for PSY 863 graduate seminar course.

GSAC Program Area Representative

Apr 2019 – Apr 2020

Psychology Graduate Student Advisory Council

Cognition and Cognitive Neuroscience (CCN) Program, Michigan State University

- Held office: Secretary
- Recorded and distributed all meeting minutes, hosted graduate student events such as annual town hall and social gatherings, conducted student elections for and trained successor.

JOURNAL EDITORIAL POSITIONS & PEER REVIEW

- Associate Editor, Illustrated Journal of Human-Computer Interaction **June 2025 - Present**
 - Ad-hoc peer-reviewer for publications: Attention, Perception, & Psychophysics; Experimental Brain Research; Frontiers in Psychology; iScience; Music Perception; Scientific Reports, Timing and Time Perception; Vision Research
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